# A New Seven-Segment Profile Algorithm for an Open Source Architecture in a Hybrid Electronic Platform

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## ABSTRACT

This paper presents a new methodology to generate a seven-segment S-curve velocity profile designed to produce smoother motion compared to traditional trapezoidal velocity profiles. By using a third-degree polynomial for position, it maintains a constant jerk value. The algorithm works with negative velocity and displacement constraints and was implemented using a hybrid electronic platform combining a SoC Raspberry Pi 3 and an FPGA.

## RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides the algorithmic basis for implementing smooth S-curve velocity profiles on the ESP32 S3 to eliminate acceleration jumps and mechanical vibrations.

## TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by José R. García-Martínez provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.